

Revision date 18-Mar-2025

Revision Number 1.1

## Section 1: Identification

### Product identifier

**Product Name** Liquichek Specialty Immunoassay Control

**Catalogue Number(s)** 359, 364, 365, 366, 359X

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** In vitro diagnostic

**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Supplier

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories Inc.  
9500 Jeronimo Road  
Irvine, California 92618  
USA

#### Importer

Bio-Rad Laboratories Pty Ltd  
189 Bush Road  
Albany Auckland  
New Zealand

**Technical Service** +64 9 415 2280 or 0508 805 500  
sales.nz@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

|                                                       |            |
|-------------------------------------------------------|------------|
| <b>Skin sensitisation</b>                             | Category 1 |
| <b>Hazardous to the aquatic environment - chronic</b> | Category 3 |

### Label elements



**Signal word**  
Warning

### **Hazard statements**

May cause an allergic skin reaction  
Harmful to aquatic life with long lasting effects

### **Precautionary Statements - Prevention**

Avoid release to the environment

Wear protective gloves, protective clothing, eye protection and face protection

**Skin**

IF ON SKIN: Wash with plenty of water and soap

If skin irritation or rash occurs: Get medical advice/attention

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other hazards which do not result in classification**

Contains animal source material. (Pig). (Cattle).

Contains human source material and / or potentially infectious components

**Section 3: Composition/information on ingredients**

| Chemical name                                                                       | CAS No.     | Weight-%     |
|-------------------------------------------------------------------------------------|-------------|--------------|
| Sodium molybdate dihydrate                                                          | 10102-40-6  | 1 - 2.5      |
| Ethyl alcohol                                                                       | 64-17-5     | 0.01 - 0.099 |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 55965-84-9  | 0.001 - 0.01 |
| Sodium azide                                                                        | 26628-22-8  | 0.001 - 0.01 |
| Non-hazardous ingredients                                                           | Proprietary | Balance      |

**Section 4: First-aid measures****Description of first aid measures**

|                       |                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                                                              |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                                                                  |
| <b>Eye contact</b>    | Contains human source material and / or potentially infectious components. Call a doctor.<br>Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a doctor.                                                     |
| <b>Ingestion</b>      | Contains human source material and / or potentially infectious components. Call a doctor.                                                                                             |

**Most important symptoms and effects, both acute and delayed**

|                            |                           |
|----------------------------|---------------------------|
| <b>Symptoms</b>            | Itching. Rashes. Hives.   |
| <b>Effects of Exposure</b> | No information available. |

**Indication of any immediate medical attention and special treatment needed**

|                        |                                                                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Note to doctors</b> | May cause sensitisation in susceptible persons. Treat symptomatically. Contains human source material and / or potentially infectious components. |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

**Section 5: Fire-fighting measures****Suitable Extinguishing Media**

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Large Fire** CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

#### **Specific hazards arising from the chemical**

**Specific hazards arising from the chemical** Product is or contains a sensitiser. May cause sensitisation by skin contact.

#### **Special protective actions for fire-fighters**

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.

### **Section 6: Accidental release measures**

#### **Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**For emergency responders** Use personal protection recommended in Section 8.

#### **Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

#### **Methods and material for containment and cleaning up**

**Methods for containment** Do not allow into any sewer, on the ground or into any body of water.

**Methods for cleaning up** Use: Disinfectant. Clean contaminated surface thoroughly.

#### **Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

### **Section 7: Handling and storage**

#### **Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse.

**General hygiene considerations** Follow universal and standard precautions for handling potentially infectious materials.

#### **Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions.

Incompatible materials                      Metals.

## Section 8: Exposure controls/personal protection

### Working area parameters, subject to mandatory control (MAC or TSEL)

#### Exposure Limits

| Chemical name                            | New Zealand                                                                                       | Australia                                        | ACGIH TLV                                                                                    | United Kingdom                                                                                       |
|------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Sodium molybdate dihydrate<br>10102-40-6 | TWA: 5 mg/m <sup>3</sup> ;                                                                        | TWA: 5 mg/m <sup>3</sup> ;                       | TWA: 0.5 mg/m <sup>3</sup> Mo<br>respirable particulate<br>matter                            | TWA: 5 mg/m <sup>3</sup> ;<br>STEL: 10 mg/m <sup>3</sup> ;                                           |
| Ethyl alcohol<br>64-17-5                 | TWA: 200 ppm;<br>TWA: 380 mg/m <sup>3</sup> ;<br>STEL: 800 ppm;<br>STEL: 1520 mg/m <sup>3</sup> ; | TWA: 1000 ppm;<br>TWA: 1880 mg/m <sup>3</sup> ;  | STEL: 1000 ppm                                                                               | TWA: 1000 ppm;<br>TWA: 1920 mg/m <sup>3</sup> ;<br>STEL: 3000 ppm;<br>STEL: 5760 mg/m <sup>3</sup> ; |
| Sodium azide<br>26628-22-8               | Ceiling: 0.11 ppm;<br>Ceiling: 0.29 mg/m <sup>3</sup> ;                                           | Peak: 0.11 ppm;<br>Peak: 0.3 mg/m <sup>3</sup> ; | Ceiling: 0.29 mg/m <sup>3</sup><br>Sodium azide<br>Ceiling: 0.11 ppm<br>Hydrazoic acid vapor | TWA: 0.1 mg/m <sup>3</sup> ;<br>STEL: 0.3 mg/m <sup>3</sup> ;<br>pSk                                 |

**Biological occupational exposure limits**                      This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

### Appropriate engineering controls

**Engineering controls**                      Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection**                      Wear safety glasses with side shields (or goggles).

**Hand protection**                      Wear suitable gloves.

**Skin and body protection**                      Wear suitable protective clothing.

**Respiratory protection**                      No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls**                      No information available.

## Section 9: Physical and chemical properties

### Information on basic physical and chemical properties

**Physical state**                      Liquid

**Appearance**                      Clear to slightly cloudy

**Colour**                      light yellow

**Odour**                      Slight.

**Odour threshold**                      No information available

| <u>Property</u>                                | <u>Values</u>     | <u>Remarks • Method</u> |
|------------------------------------------------|-------------------|-------------------------|
| <b>pH</b>                                      | 6.8-7.6           | None known              |
| <b>Melting point / freezing point</b>          | No data available | None known              |
| <b>Initial boiling point and boiling range</b> | No data available | None known              |
| <b>Flash point</b>                             | No data available | None known              |

|                                        |                           |            |
|----------------------------------------|---------------------------|------------|
| Evaporation rate                       | No data available         | None known |
| Flammability                           | No data available         | None known |
| Flammability Limit in Air              |                           | None known |
| Upper flammability or explosive limits | No data available         |            |
| Lower flammability or explosive limits | No data available         |            |
| Vapour pressure                        | No data available         | None known |
| Relative vapour density                | No data available         | None known |
| Relative density                       | No data available         | None known |
| Water solubility                       | Miscible in water         |            |
| Solubility(ies)                        | No data available         | None known |
| Partition coefficient                  | No data available         | None known |
| Autoignition temperature               | No data available         | None known |
| Decomposition temperature              |                           | None known |
| Kinematic viscosity                    | No data available         | None known |
| Dynamic viscosity                      | No data available         | None known |
| Explosive properties                   | No information available. |            |
| Oxidising properties                   | No information available. |            |
| <b>Other information</b>               |                           |            |
| Softening point                        | No information available  |            |
| Molecular weight                       | No information available  |            |
| VOC content                            | No information available  |            |
| Liquid Density                         | No information available  |            |
| Bulk density                           | No information available  |            |
| Particle characteristics               | No information available  |            |

## Section 10: Stability and reactivity

### Reactivity

**Reactivity** No information available.

### Chemical stability

**Stability** Stable under normal conditions.

### Explosion data

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

### Possibility of hazardous reactions

**Possibility of hazardous reactions** Avoid contact with metals. This product contains Sodium azide. Sodium azide can react with Copper, Brass, Lead, and solder in piping systems to form explosive compounds and toxic gases.

### Conditions to avoid

**Conditions to avoid** None known based on information supplied.

### Incompatible materials

**Incompatible materials** Metals.

### Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

## Section 11: Toxicological information

### Acute toxicity

### Information on likely routes of exposure

#### Product Information

|                     |                                                                                                                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available.                                                                                                                                                          |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available.                                                                                                                                                          |
| <b>Skin contact</b> | May cause sensitisation by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available.                                                                                                                                                          |
| <b>Symptoms</b>     | Itching. Rashes. Hives.                                                                                                                                                                                                    |

### Acute toxicity

#### Numerical measures of toxicity

The following ATE values have been calculated for the mixture  
 ATE<sub>mix</sub> (oral) 71,608.50 mg/kg

#### Component Information

| Chemical name                                                                          | Oral LD50            | Dermal LD50              | Inhalation LC50                                      |
|----------------------------------------------------------------------------------------|----------------------|--------------------------|------------------------------------------------------|
| Ethyl alcohol                                                                          | = 7060 mg/kg ( Rat ) | -                        | = 116.9 mg/L ( Rat ) 4 h<br>= 133.8 mg/L ( Rat ) 4 h |
| 5-Chloro-2-methyl-3(2H)-isothiazolone,<br>mixture with<br>2-methyl-3(2H)-isothiazolone | = 53 mg/kg ( Rat )   | = 87.12 mg/kg ( Rabbit ) | -                                                    |
| Sodium azide                                                                           | = 27 mg/kg ( Rat )   | = 20 mg/kg ( Rabbit )    | 0.054 - 0.52 mg/L ( Rat ) 4 h                        |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                      |
|------------------------------------------|--------------------------------------|
| <b>Skin corrosion/irritation</b>         | No information available.            |
| <b>Serious eye damage/eye irritation</b> | No information available.            |
| <b>Respiratory or skin sensitisation</b> | May cause an allergic skin reaction. |
| <b>Germ cell mutagenicity</b>            | No information available.            |

#### Carcinogenicity

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name           | New Zealand | IARC    |
|-------------------------|-------------|---------|
| Ethyl alcohol - 64-17-5 | -           | Group 1 |

#### Legend

IARC (International Agency for Research on Cancer)

Group 1 - Carcinogenic to Humans

Reproductive toxicity No information available.

STOT - single exposure No information available.

STOT - repeated exposure No information available.

Aspiration hazard No information available.

Data used to identify the health effects Refer to Section 16 for Key literature references and sources for data used to compile the SDS.

**Section 12: Ecological information****Ecotoxicity**

Aquatic ecotoxicity Harmful to aquatic life with long lasting effects.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                | Crustacea                                                                        |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Ethyl alcohol | -                    | LC50: 12.0 - 16.0mL/L (96h, Oncorhynchus mykiss)<br>LC50: >100mg/L (96h, Pimephales promelas)<br>LC50: 13400 - 15100mg/L (96h, Pimephales promelas) | LC50: 9268 - 14221mg/L (48h, Daphnia magna)<br>EC50: =2mg/L (48h, Daphnia magna) |
| Sodium azide  | -                    | LC50: =0.8mg/L (96h, Oncorhynchus mykiss)<br>LC50: =0.7mg/L (96h, Lepomis macrochirus)<br>LC50: =5.46mg/L (96h, Pimephales promelas)                | -                                                                                |

**Terrestrial ecotoxicity**

| Chemical name | Earthworm                                                                            | Avian | Honeybees |
|---------------|--------------------------------------------------------------------------------------|-------|-----------|
| Ethyl alcohol | Acute Toxicity: LC50 0.1 - 1 mg/cm <sup>2</sup> (Eisenia foetida, 48 h filter paper) | -     | -         |

Persistence and degradability No information available.

**Bioaccumulative potential****Bioaccumulation****Component Information**

| Chemical name                                                                    | Partition coefficient |
|----------------------------------------------------------------------------------|-----------------------|
| Ethyl alcohol                                                                    | -0.35                 |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | 0.7                   |

**Mobility in soil**

**Mobility** No information available.

**Other adverse effects**

No information available.

**Section 13: Disposal considerations****Disposal methods****Waste from residues/unused products**

Dispose of product in packaging in a way that is consistent with the EPA Consolidation 30 April 2021 of the Hazardous Substances (Disposal) Notice 2017 and the Act.

Treat the substance using a method that changes the characteristics or composition of the substance so that the substance is no longer a hazardous substance; or export the substance from New Zealand as waste.

Substances which are hazardous to human health or corrosive to metals – may be discharged into the environment if a tolerable exposure limit has been set for the substance (or a component of that substance); and the discharge does not, after reasonable mixing, result in the concentration of the substance in an environmental medium exceeding the tolerable exposure limit. If there is no tolerable exposure limit for the substance, then it may only be discharged into the environment if the substance is very rapidly converted to substances that are not hazardous substances.

Environmentally hazardous substances – if the substance, or if it contains a component that is hazardous to the aquatic environment or bioaccumulative and not rapidly degradable, then any component that is bioaccumulative and not rapidly degradable must be removed.

The product may only be discharged into the environment if an environmental exposure limit has been set for the substance (or a component of the substance); and the discharge does not, after reasonable mixing, result in the concentration of the substance in an environmental medium exceeding the environmental exposure limit.

Dispose of in accordance with local regulations.

Dispose of waste in accordance with environmental legislation.

Flush pipes with water frequently if discarding solutions containing Sodium azide into metal piping systems.

**Contaminated packaging**

For packages that have been in direct contact with hazardous substances, the person must ensure that the package is rendered incapable of containing any substance. It must be disposed of in a manner that is consistent with the requirements for disposal of the substance that it contained, taking into account the material the package is manufactured from.

Packages may only be reused or recycled if:

- the substance has a physical hazard other than corrosive to metal, and has been treated to remove any residual contents of the hazardous substance;
- or for substances that have a health or environmental hazard, or corrosive to metal, the contents of the residue in the package are below the threshold for the substance to be classified as hazardous in the Hazardous Substances (Hazard Classification) Notice 2020.

**Section 14: Transport information**

**IATA** Not regulated

**IMDG** Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

**Special precautions for user**

Please refer to the applicable dangerous goods regulations for additional information



## Section 15: Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

**EPA New Zealand HSNO approval code or group standard** To be determined

#### **National regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### **Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information

Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information

Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

## Section 16: Other information

**Revision date** 18-Mar-2025

**Revision Note** Reviewed existing information and made minor updates.

### **Key or legend to abbreviations and acronyms used in the safety data sheet**

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)

New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**